Changelog

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This document records the revisions made to Deliverable 1 in response to the feedback received on the first submission (version dated 4 June 2026). Entries are grouped under the headings used in that feedback. The revised document is written under the extended allowance of 2 000 words granted with the feedback.

Feedback point	**Change made**	**Where**
Domain Description before Purpose and Scope	Section order inverted.	Domain Description
Intended Audience shortened or integrated	Standalone section removed; content condensed into one paragraph.	Domain Description, final paragraph
Significance and Impact after the project outline	Moved to last body section.	Significance and Impact
Dedicated Data section	Added; signals and datasets consolidated from multiple sections.	Data and Evaluation
State the intended mathematical formulation	New section: tokenisation operator, phase-locking conditions, locked frequency class, operational definition of structural aliasing.	Mathematical Formulation, Eqs. (1)–(8)
Reformulate the hypotheses explicitly	Objectives replaced by H1–H3 with predicted direction, metric and refutation criterion.	Aliasing implications
Make the probing experiments precise	New section: probed representations, metric, probe points and controls.	Probing Methodology
Simplify the experimental pipeline	Staged: default checkpoint on sinusoids $\rightarrow$ retrained (P, S) grid $\rightarrow$ complex synthetics $\rightarrow$ photovoltaic data.	Data and Evaluation; Methodology
Clarify whether stride changes require retraining	Retraining necessity shown for both P and S; recipe and departures tabulated.	Methodology, Tables 1–2
Explore strides on both sides of the default	Admissibility constraint $S \leq P$ stated; symmetric design built on the $P{=}S$ diagonal: $\{8, 16, 24\}$.	Methodology
Describe the planned priors	Likelihoods, link function and all priors stated with justification.	Bayesian Analysis, Eqs. (9)–(10)

Table 1: *Summary of the feedback and corresponding revisions.*

During the preparation of this work, the authors used AI-based tools, including GPT-5.5, Gemini 3.6 Flash, Sonnet 5, and Opus 4.6, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

Structure and organisation

The document now opens with *Domain Description*, followed by *Purpose and Scope*. The standalone *Intended Audience* section was condensed into a closing paragraph of *Domain Description*, and *Significance and Impact* was moved to the end of the body. The single *Project Outline* section was replaced by dedicated sections (formalisation, data, methodology, probing, statistics, risk assessment), and a *Data and Evaluation* section consolidates all signals and datasets previously spread across multiple sections. The ontology was moved to Appendix A.

Mathematical formulation and hypotheses

A *Mathematical Formulation* section was added, where the first submission only announced it as planned work. It defines the tokenisation operator $\mathcal{T}_{P,S}$ (Eq. 1), the signal model with Nyquist constraint (Eq. 2), the full stride-translation derivation (Eqs. 3–5), the analogous patch-length condition (Eq. 6), and the phase-locking frequency class $\mathcal{F}_{\text{lock}}$ as a function of f_s, P and S (Eq. 7). It is now stated that patch self-repetition does not by itself imply that two distinct signals become indistinguishable; structural aliasing is given a separate operational definition at the representation level (Eq. 8). The three former objectives were reformulated as hypotheses H1–H3, each with a predicted direction, the relevant metric, and an explicit refutation criterion. The ontology is set out in Appendix A; the generation-state axioms are given in full, while the remaining axiom families are outlined for feedback before formalisation.

Experimental methodology

A Probing Methodology section replaces the single sentence of the first submission, specifying the probed representations, the MDL-based metric with its controls, and the per-frequency accuracy profile. Probing was extended from decoder-only to include the encoder, so that the stage at which frequency information is lost can be localised. The pipeline was simplified and staged: clean sinusoids first, then the retrained (P, S) grid, then the photovoltaic application.

Patch stride experiments

It is now stated that changing P or S requires retraining (the former resizes the patch-embedding block, the latter alters inter-patch dynamics). Table 1 reports the five retrained configurations; Table 2 compares all hyperparameters against the published Chronos recipe and isolates the departures. The unpublished Chronos-Bolt recipe is approximated with the published Chronos-T5 one, under a uniform 100k-step budget to avoid confounding. Configurations with $S > P$ are excluded: they discard samples between patches, confounding structural aliasing with data loss. The requested symmetry is realised on the $P = S$ diagonal ($\{8, 16, 24\}$), the only axis admitting neighbours on both sides under $S \leq P$.

The context window was corrected to $T = 2048$ samples (the first submission reported $T = 512$ tokens). The exclusion of $S = 4$ is now justified. Figure 1 is included as evidence that the configurations train to a usable state; it is not offered as analysis, which belongs to later deliverables.

A mitigation strategy and its link to agent behaviour were also added: reducing S shrinks $\mathcal{F}_{\text{lock}}$, and the `pv:AliasingEvent` class lets the control agent flag reduced confidence at locked frequencies.

Bayesian analysis

A *Bayesian Analysis* section was added. It splits the analysis into a relative part (logarithmic local contrast d pairing each locked frequency with matched controls, Eq. 9, under a heavy-tailed Student-t_4 likelihood with prior $\bar{\beta} \sim \text{Student-}t_4(0, 0.5)$) and an absolute part (Gamma regression with log link on MDL codelength, Eq. 10, with priors on θ_{lock} and the shape parameter). Planned inference: 95% credible intervals and posterior probability of ≥20% attenuation.